

SAGAR KHAIRE

● sagarkhaire456@gmail.com ● (361)228-6684 ● www.linkedin.com/in/sagar-khaire
● Detroit, MI

Dedicated, hardworking and an energetic mechanical professional with relevant technical know-how and skills in the field of product development. A stickler for precision and detail and a highly competent in challenging work environment. Looking for an opportunity as a mechanical engineer in an esteemed organization to achieve career objectives.

Skills

- 2D & 3D CAD SolidWorks, AutoCAD, CATIA
- CAE ANSYS (certified by IGTR)
- Product manufacturing Sheet metal, Injection molding, casting, forging, GD&T
- Lean Manufacturing 5S, Kaizen, Time study
- Quality Engineering PPAP, DFMEA, PFMEA, APQP
- Microsoft office

Education

MS in Mechanical Engineering	Texas A&M University	GPA 3.4/4.0	Dec 2016
BE in Mechanical Engineering	University of Pune	GPA 3.0/4.0	May 2015

Professional Experience

Mechanical Design Engineer Delray Mechanical Corp. Apr 2017-present

- Responsible for **DFM** and product specifications addressing customer requirements using SolidWorks
- Applied modeling principles to heavy machine components resulting into **material optimization by 30%**
- Achieved **production time reduction** by generating sheet metal design templates
- Implemented **GD&T** as per ASME Y14.5 for creating mechanical drawings and **BOMs**
- Accountable for **design release, FMEA, lean manufacturing, on-site technical support and technical supervision**

Product Engineer Patson Machines Pvt. Ltd Nov 2014-Jun 2015

- Designed and manufactured a portable drilling unit for a **Special Purpose Machine (SPM)**
- Achieved **better precision and accuracy** by **overcoming the drawbacks of conventional methods**
- Generated mechanical designs in **AutoCAD and CATIA** followed by **FEA simulation** using **ANSYS**
- **Manufactured the components required set of tools** using manual manufacturing processes
- Executed **lean manufacturing** techniques such as **5S, Kaizen and Six Sigma**

Trainee Engineer BharatForge Pvt. Ltd. May 2015-Jul 2015

- Created 2D/3D part designs for powertrain components using AutoCAD
- Gained hands on working experience with die casting, forging and injection molding processes
- Performed Non-Destructive Testing (NDT) and prepared reports

Academic Experience

Finite Element Analysis of fuselage of Boeing 737 Jan 2016-May 2016

- Designed the **fuselage using SolidWorks** as per forces and aerodynamic dispositions
- Conducted appropriate **material selection** from the results of **stress analysis** simulation on **ANSYS**

Formation of a Cubic Bazier Curve using Computer Aided Geometric Design (CAGD) Jun 2016-Jul 2016

- Developed a cubic bazier curve utilizing the divide and conquer approach
- Performed hand calculations for designing an algorithm and a **MATLAB** code was developed based on the algorithm

Research and Term Paper on Polymer Additives in Papermaking Aug 2016-Dec 2016

- Explored the concept of **polymer additives to increase the tensile strength with weight optimization**
- Studied polymer additives with their specifications and **effects on conventional papermaking process**
- Combined effects of the additives were plotted to achieve the objective

Jet Propulsion in Turbojet Engine Jan 2016-May 2016

- Formulated the mathematical **relation between developed thrust and exit area of nozzle**
- Utilized concluded results to **model and predict the engine type with increase in engine performance**
- **Performed thrust and dimensional calculations** using propulsion system parameters and plotted the results